E-ArtTech

Electronic Art Technology

WORKSHOP ON

PCB DESIGNING

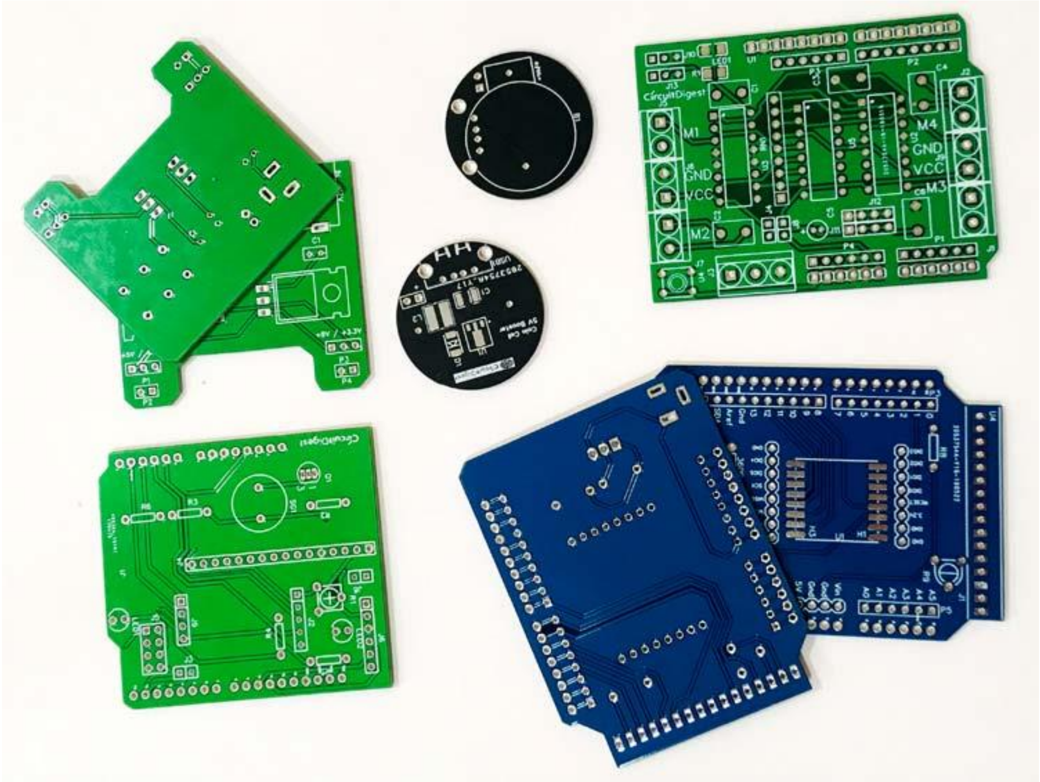
&

FABRICATION

Design By :-Mr. Kunal Khawashi

What is PCB?

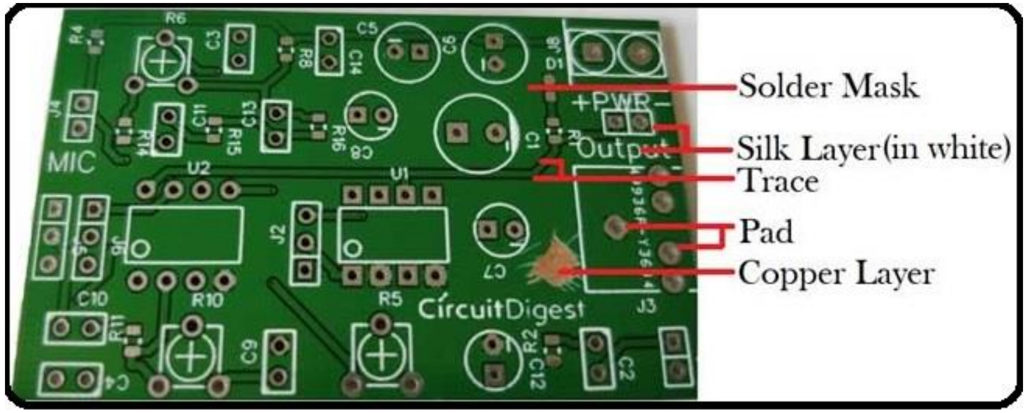
- PCB or Printed Circuit Board is the traditional name for the bare board of which you supply us with the layout data and which you use to mount your components.
- is used to mechanically support and electrically connect electronic components using conductive pathways, tracks or signal traces etched from copper sheets laminated onto a non-conductive substrate.
- A PCB populated with electronic components is called a printed circuit assembly (PCA)



Materials of PCB

PCB generally consists of four layers, which are heat laminated together into a single layer.

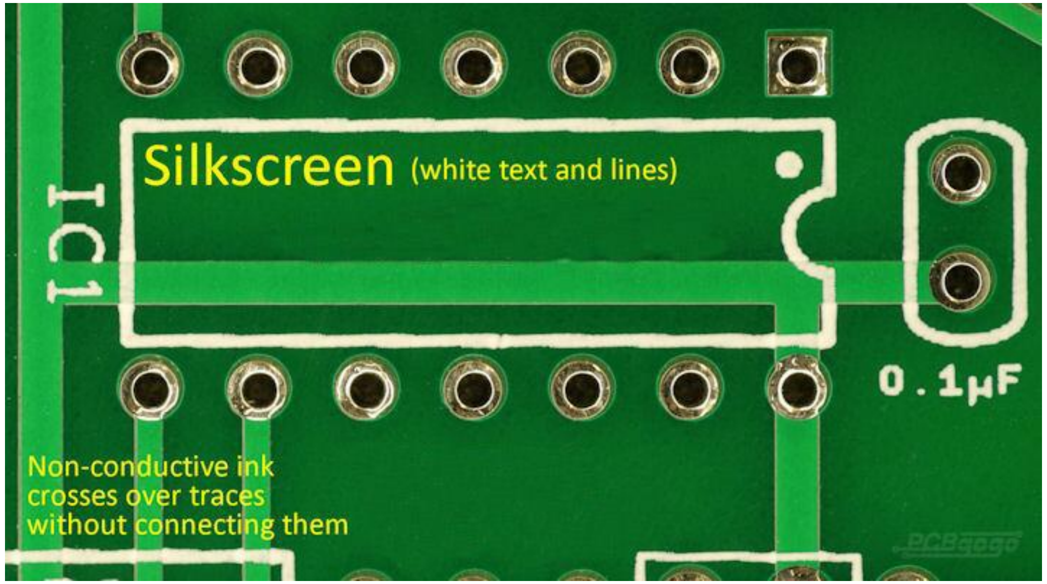
The different types of PCB materials used in PCB from top to bottom includes



- **Silk screen**
- **Solder mask**
- **Copper Foil**
- **Substrate/Core**

Silk layer

- Silkscreen is a layer of ink traces used to identify components, test points, parts of the PCB, warning symbols, logos and marks etc.
- This silkscreen is usually applied on the component side.
- silkscreen can help both the manufacturer and the engineer to locate and identify all the components.
- The ink is a non-conductive epoxy ink.
- Silk layer can be utilized as a part of best or potentially base layer of PCB as indicated by client necessity which is known as TOP silk screen and BOTTOM silk screen.



Solder mask

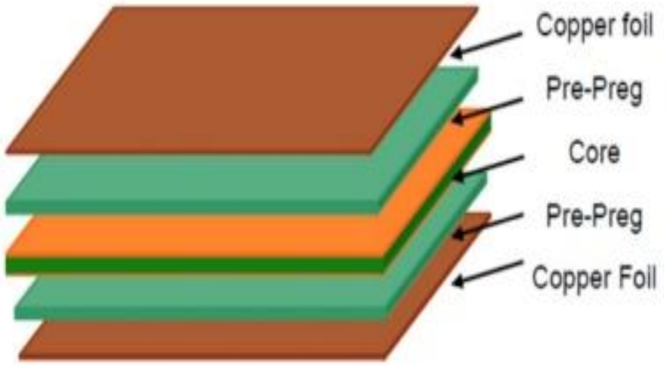
- After PCB fabrication, the copper traces on the board typically face the risk of oxidation and corrosion due to environmental exposure.
- The most reliable way to prevent this and increase the life of the PCB is to provide a protective coating known as a solder mask layer.
- Prevents solder form binding between conductors and thereby creating short circuits
- Designed to keep solder only in certain areas

Copper Foil

- In PCB industry, it exists as copper foil in rolls, which can be pressed together with prepreg to the rigid or flexible copper clad laminate under heat and pressure, then to be etched into conductors of heat and electricity on conductive layers.

- PCB copper foil is the initial copper thickness applied on outer and inner layers of a multilayer PCB board.

- Copper foil has lower rate of surface oxygen and can be pre-attached by laminate manufacturers to various base materials, such as metal core, polyimide, FR-4, PTFE and ceramic, to produce copper clad laminates.



Substrate/Core

- The PCB substrate is the physical material that holds the traces and components.
- Is made of fiberglass and is also known as FR4, with the FR letters standing for "fire retardant“.
- The massively-used copper clad laminate (CCL) can be classified into different categories:-

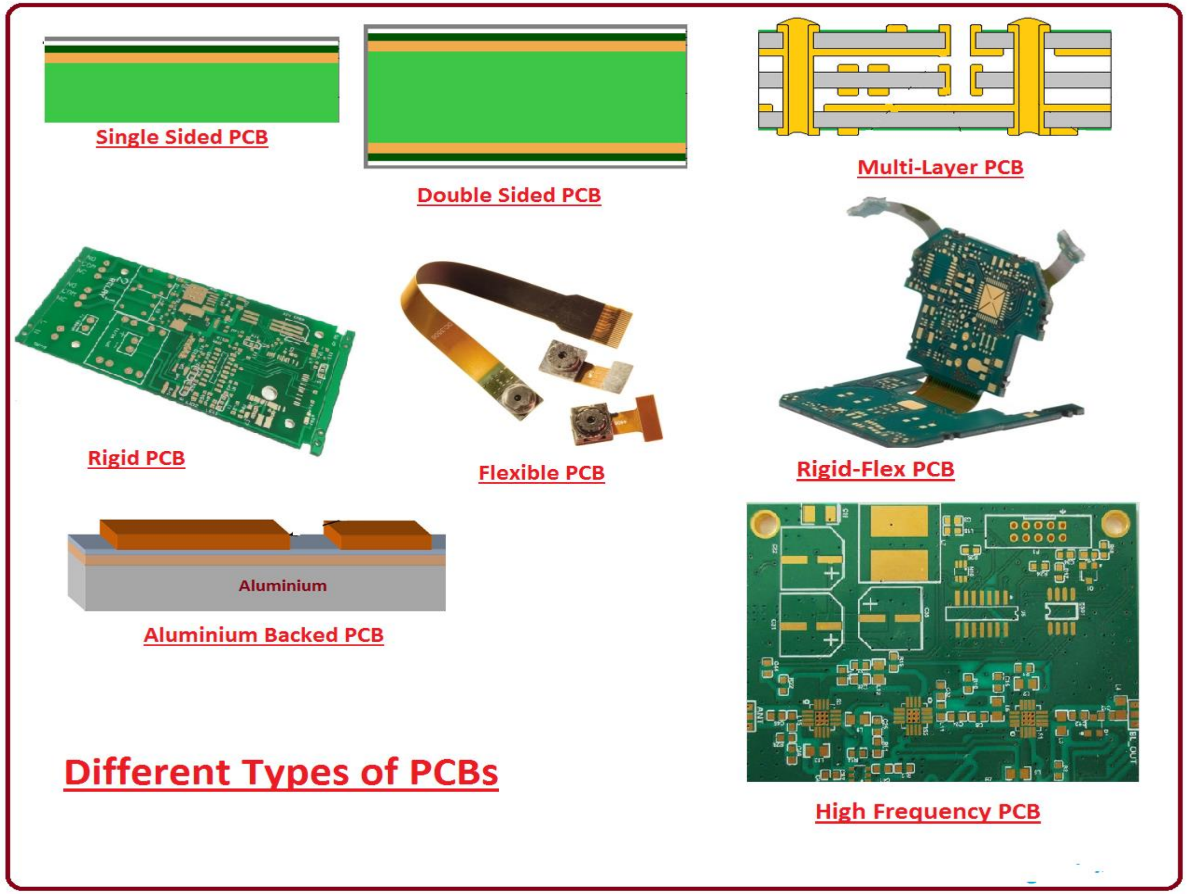
➢ Reinforcing material

➢ flammability

Classification Standard		Material
Reinforcing Material	Paper base class	PF resin (XPC, FR1, FR2) Epoxy resin (FE-3) Polyester resin
	Glass fiber cloth base class	Epoxy resin (FR4, FR5)
	Composite epoxy material (CEM)	/
	Lamination multilayer base class	/
	Special material base class	BT, PI, PPO, MS
Flammability	Flame-proof type	UL94-VO, UL94-V1
	Non-flameproof type	UL-94-HB
CCL Performance	CCL with ordinary performance	/
	CCL with low dielectric constant	/
	CCL with high heat resistance	/
	CCL with low coefficient of thermal expansion	/

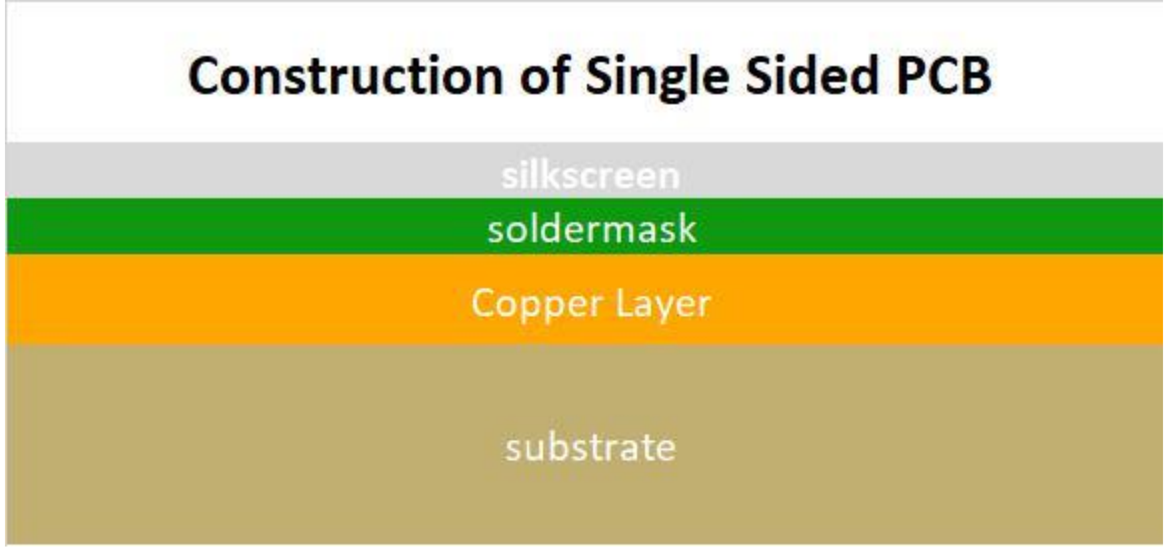
Common Types of Printed Circuit Boards

- Single Layer PCB
- Double Layer PCB
- Multi-Layer PCB
- Rigid Circuit Board
- Flex PCB | Flexible PCB
- Metal Core PCB
- Rigid Flex PCB



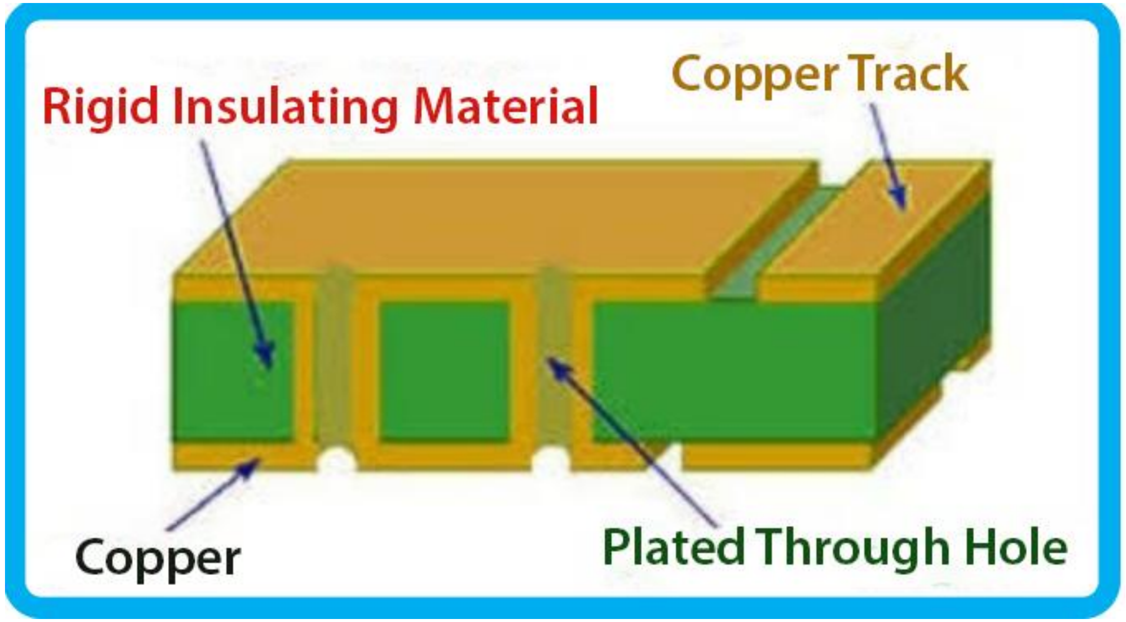
Single Layer PCB

- Single layer printed circuit boards are among some of the simplest to design and manufacture.
- These boards have a single layer of conductive material (such as copper) on only one side of a non-conductive substrate.



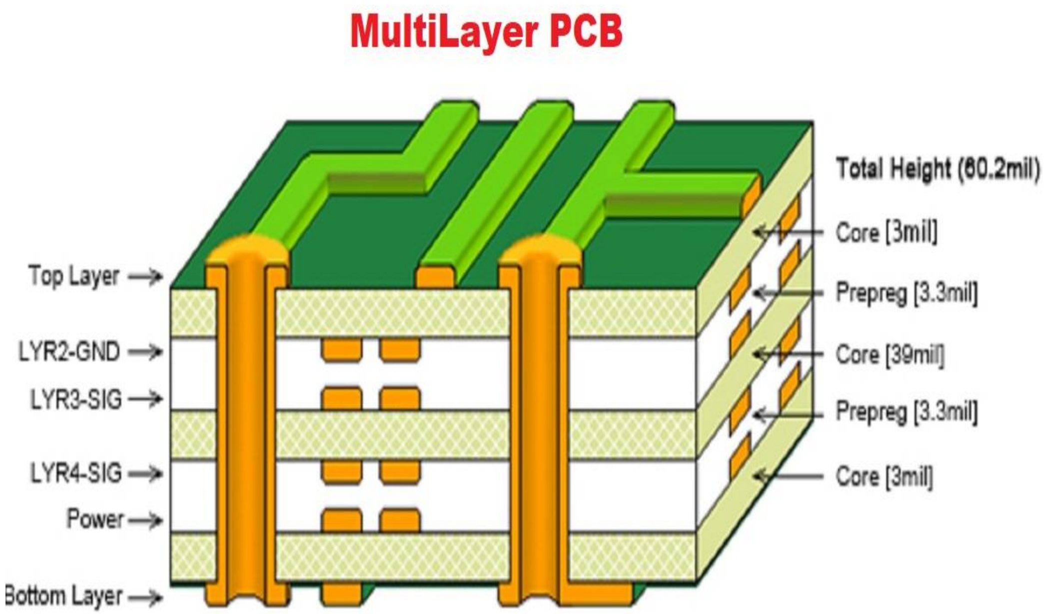
Double Layer PCB

- Double layer PCBs have one conductive layer on top of a non-conductive substrate and another conductive layer on the reverse side (top and bottom layers).
- The two conductive sides can be connected using plated holes in the substrate that connect to pads on both sides of the circuit board; these are called vias.



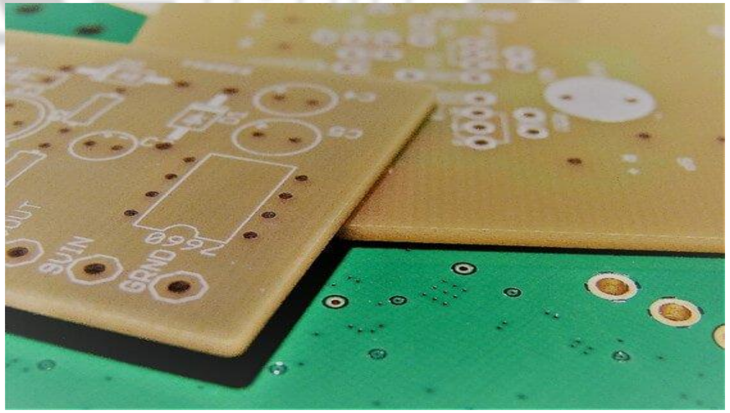
Multi-Layer PCB

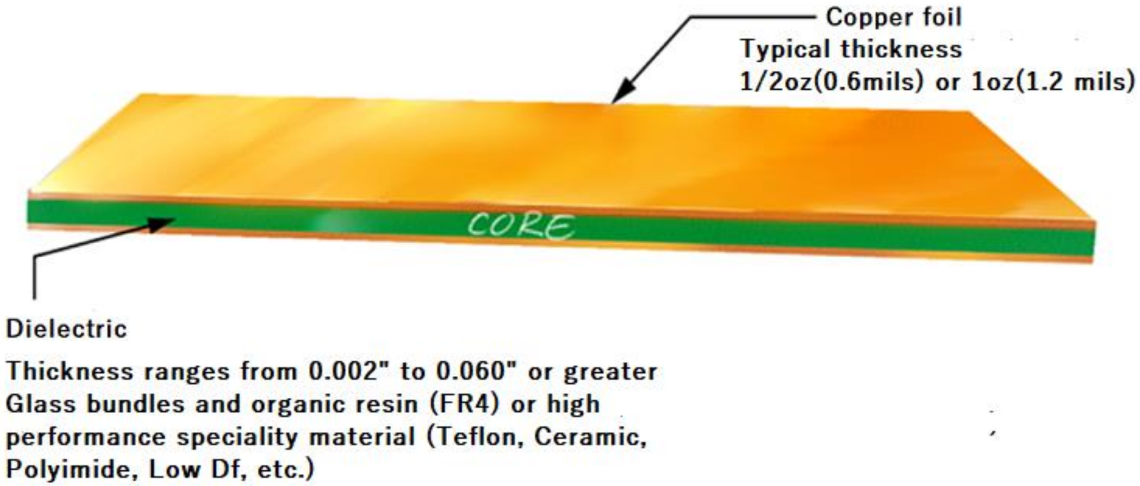
- This term refers to a circuit board that has three or more conductive layers.
- The conductive layers are at the top and bottom, as well as at least one conductive layer sandwiched between non-conductive substrate.
- Advanced Circuits has the capability to fabricate up to 40 layer boards, but the most commonly used have lower layer counts such as 4, 6, or 8 layer boards.



Rigid Circuit Board

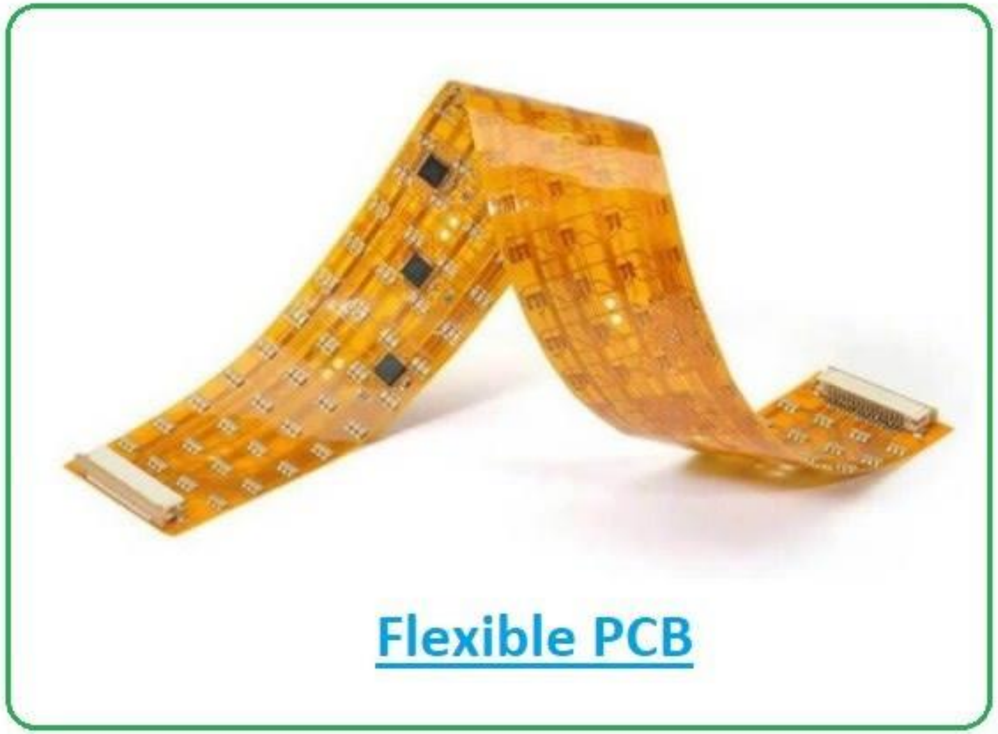
- Rigid Circuit Board is a Board that we cannot bend or force out of shape. It is not flexible.
- A Rigid PWB can be Single Sided, Double Sided or Multilayer.
- Once a Rigid Circuit Board is manufactured they cannot be modified or folded into any other shape.





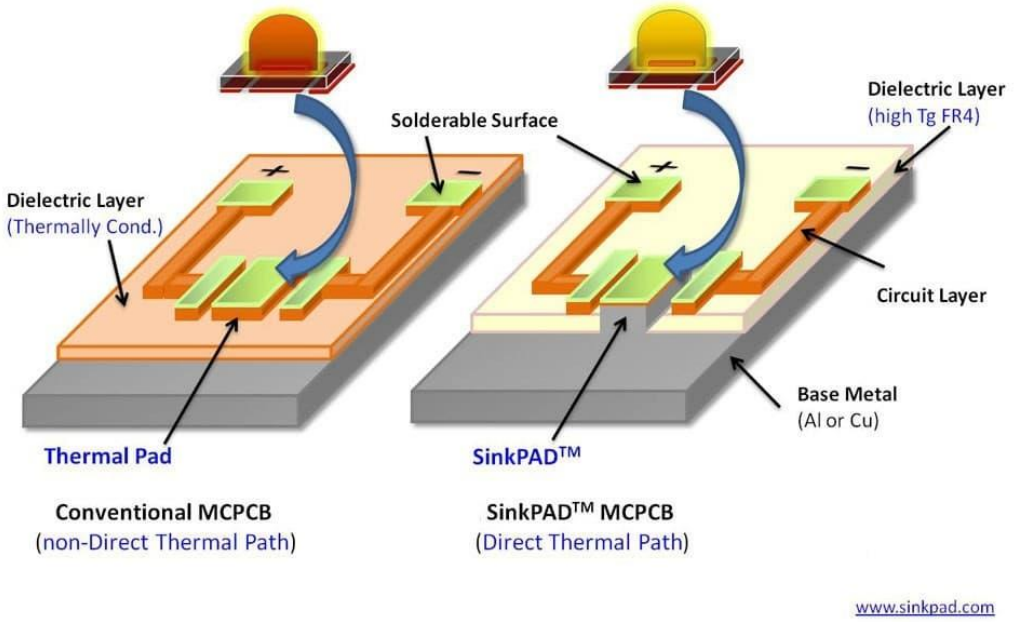
Flex PCB | Flexible PCB

- Flex PCB or Flexible PCB or Flex circuit or Flexi Circuits are terms used for a Flexible Printed Circuit Board.
- Flex PCB are flexible and can be folded and are not hard as Rigid PCB.
- A Flex PCB can be single sided PCB, double-sided PCB or multilayer PCB.
- The substrate of a flexible board is made of flexible plastic (thin insulating polymer film), polyimide or a similar polymer or Kapton.
- Generally SMD Components *are* soldered on a Flexible PCB.



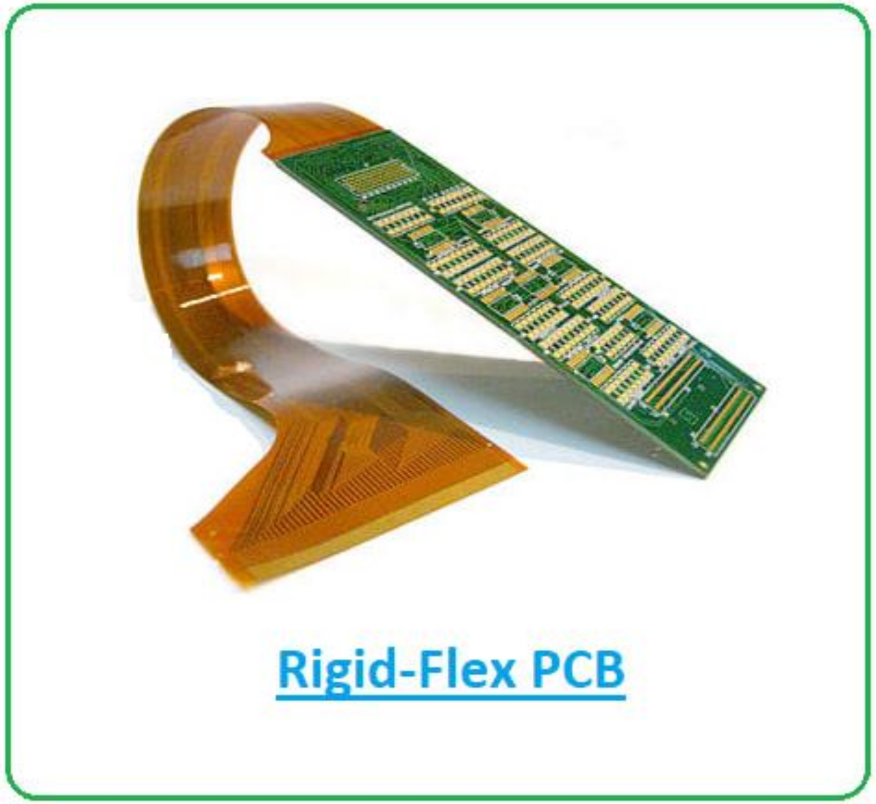
Metal Core PCB

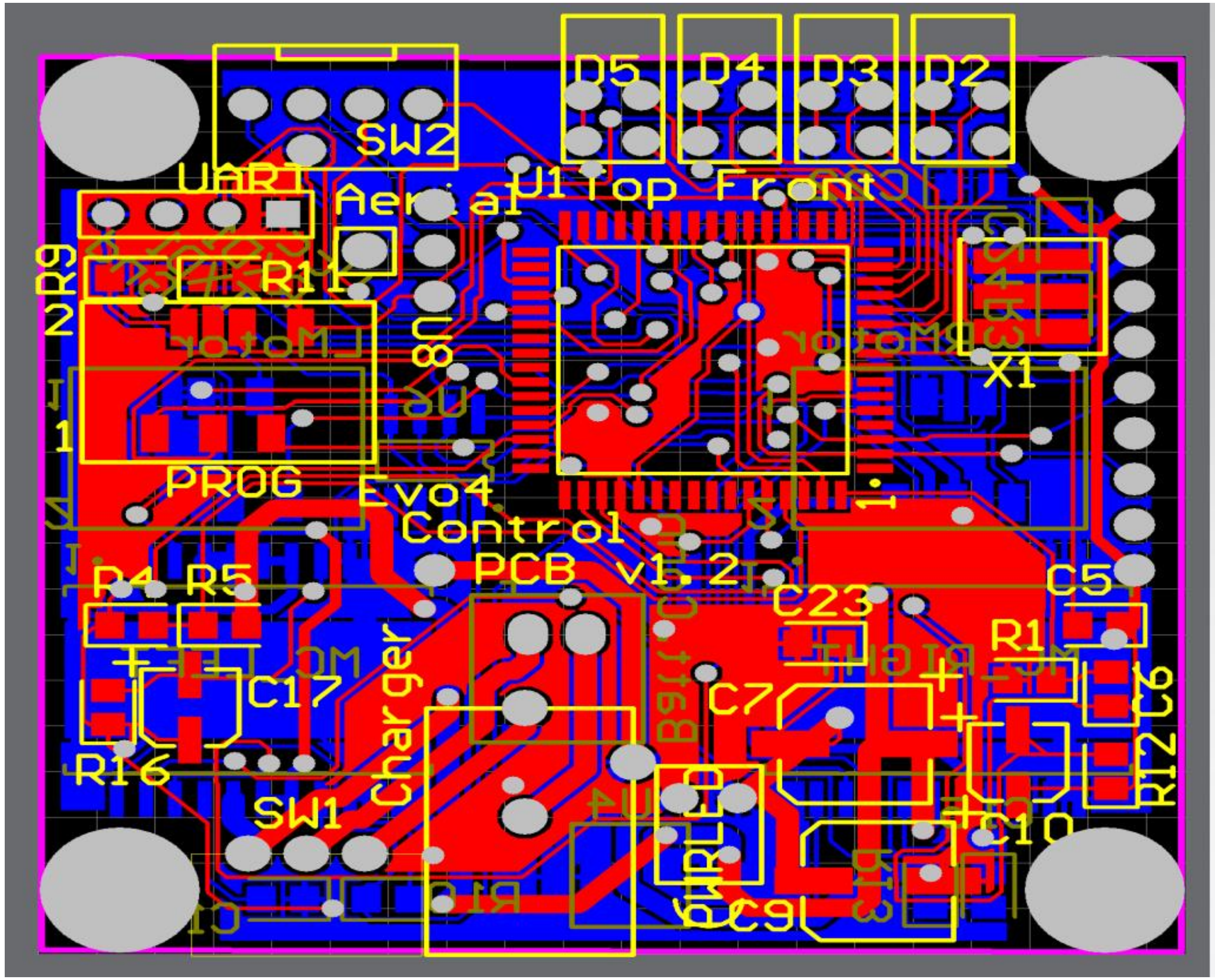
- A Metal Core Printed Circuit Board (MCPCB), also known as a thermal PCB or metal backed PCB, is a type of PCB that has a metal material as its base for the heat spreader portion of the board.
- The thick metal (almost always aluminum or copper) is covering 1 side of the PCB. Metal core can be in reference to the metal, being either in the middle somewhere or on the back of the board.
- The purpose of the core of an MCPCB is to redirect heat away from critical board components and to less crucial areas such as the metal heatsink backing or metallic core.

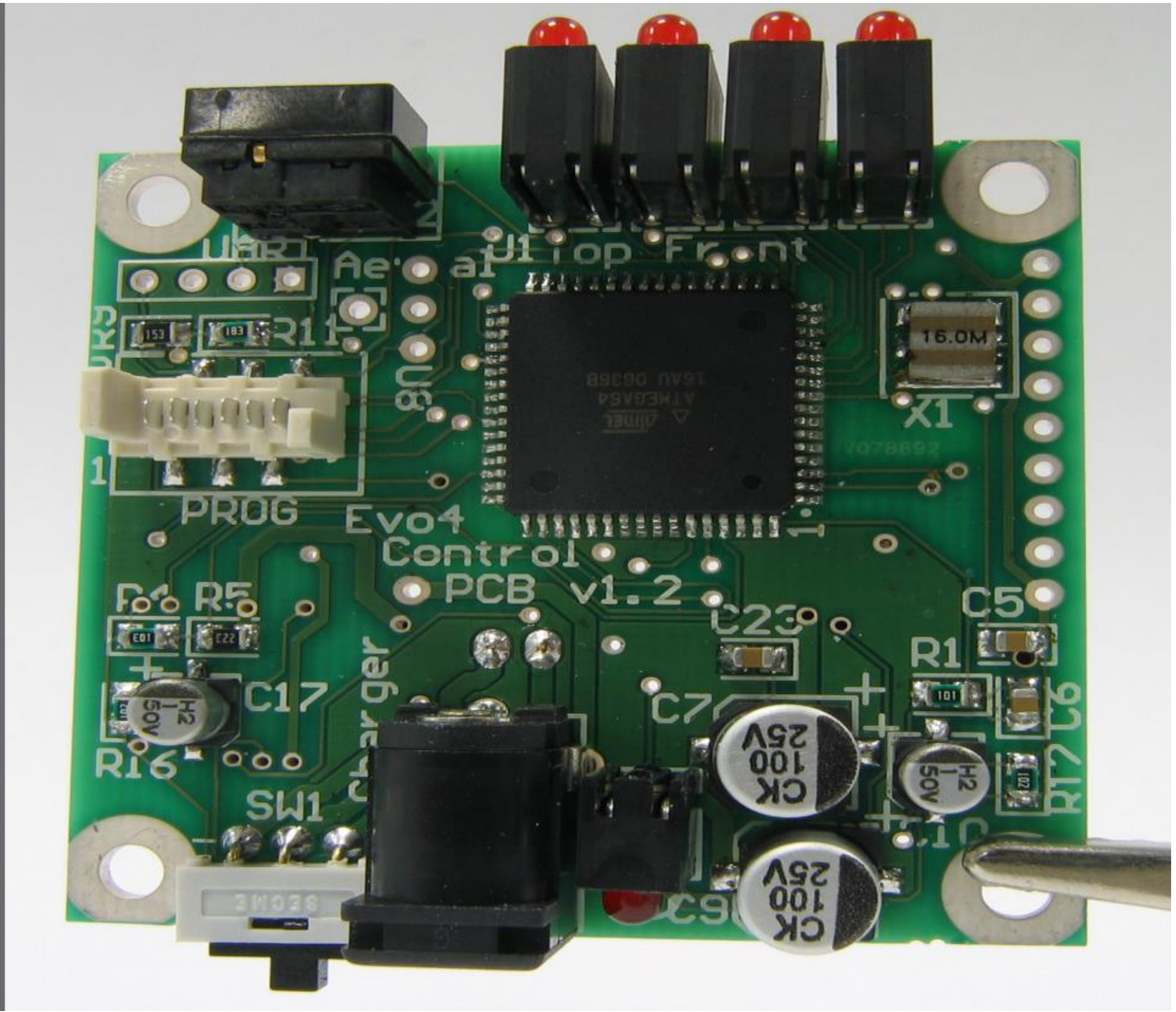


Rigid Flex PCB

- Rigid flex PCB is a hybrid combination of a Rigid PCB and a Flex PCB.
- Rigid flex circuit boards consist of both rigid and flexible substrates that are laminated together to form a single circuit board.
- Double sided or multilayer Rigid Flex Circuit Boards are interconnected by plated through holes (*PTH*).




D5
D4
D3
D2
SW2
UART
Aerial
U1 Top Front
R9
R11
PROG
Evo4
Control
PCB v1.2
X1
R4
R5
C23
C5
R1
C17
Charger
C7
C6
R12
R16
SW1
PWRLED
C10
C9


UART
Aerial
U1 Top Front
R11
16.0M
X1
PROG
Evo4
Control
PCB v1.2
R4
R5
C23
C5
R1
C17
Charger
C7
R16
SW1
C6
R12
C10
C9
CK 100 25V

Parts of a PCB

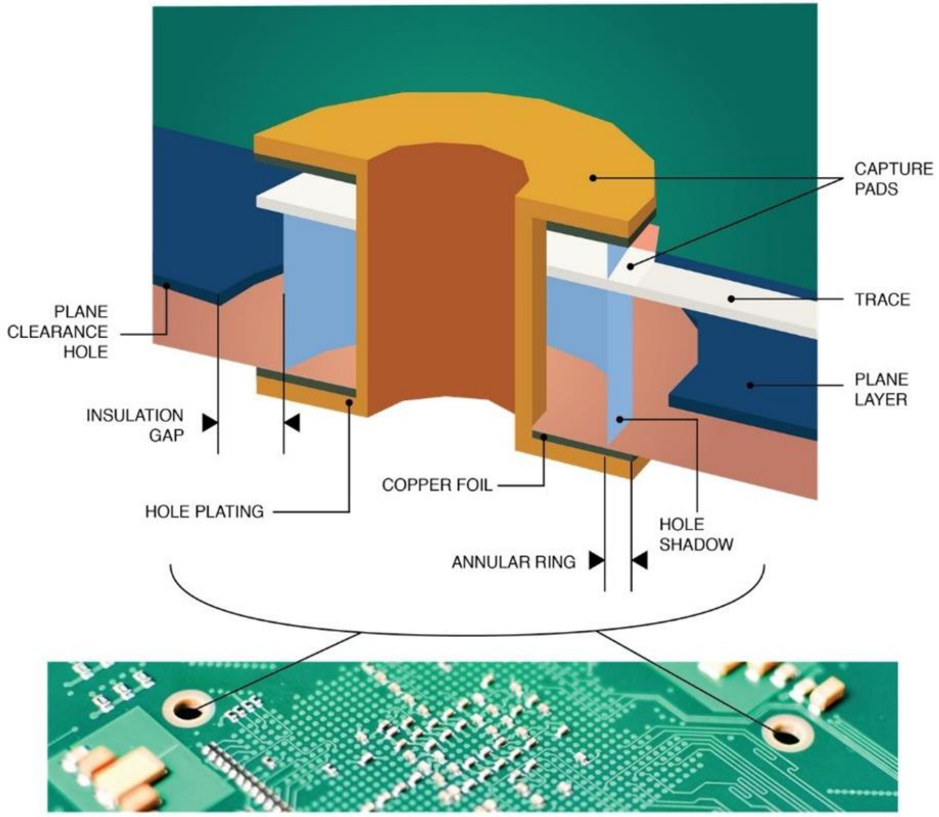
- Components
- Pads
- Traces
- Vias
- Top Metal Layer
- Bottom Metal Layer

Components

- Components are the actual devices used in the circuit.
- This includes input/output connections.
- I/O ports, including power supply connections, are also important in the PCB design.

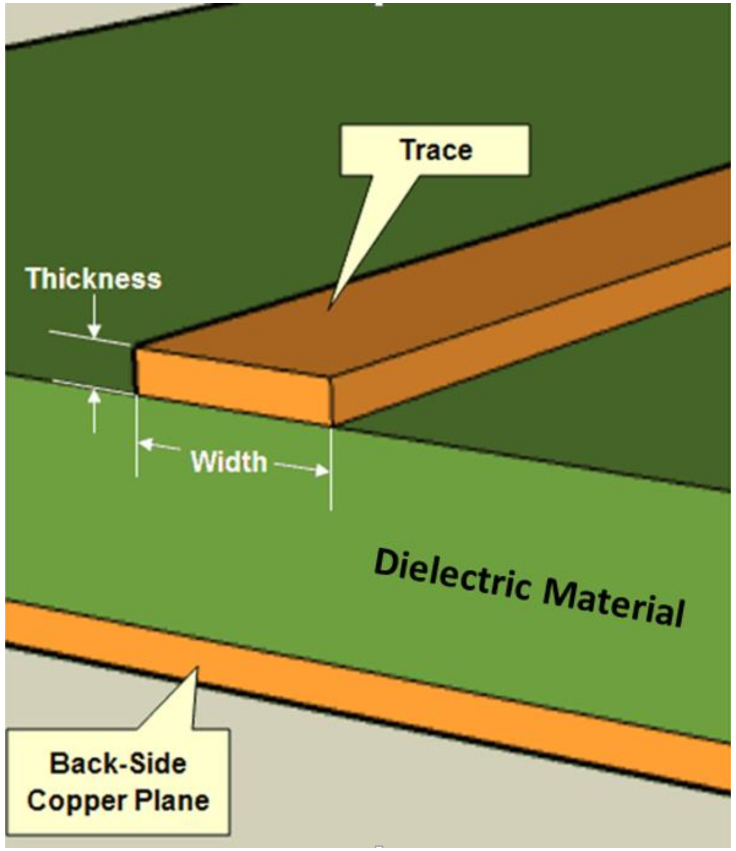
Pads

- Location that components connect to.
- You will solder components to the pads on the PCB.
- Pads will connect to traces.
- Pads have an inner diameter and outer diameter.



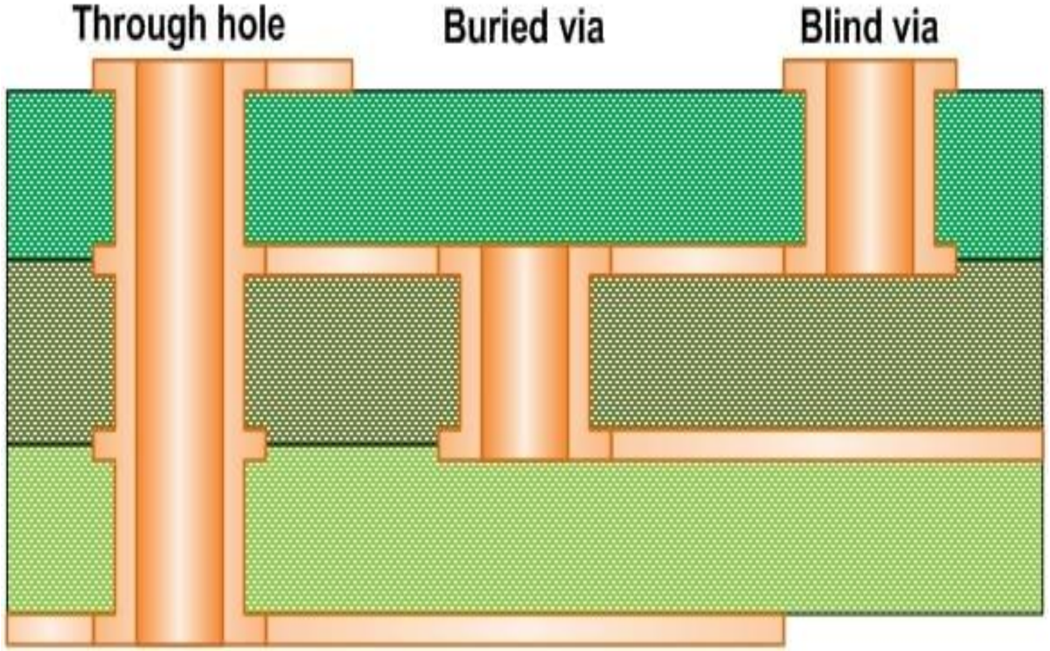
Traces

- Traces connect pads together.
- Traces are essentially the wiring of the PCB.
- Equivalent to wire for conducting signals
- Traces sometimes connect to vias.
- High current traces should be wide.
- Signal traces usually narrower than power or ground traces



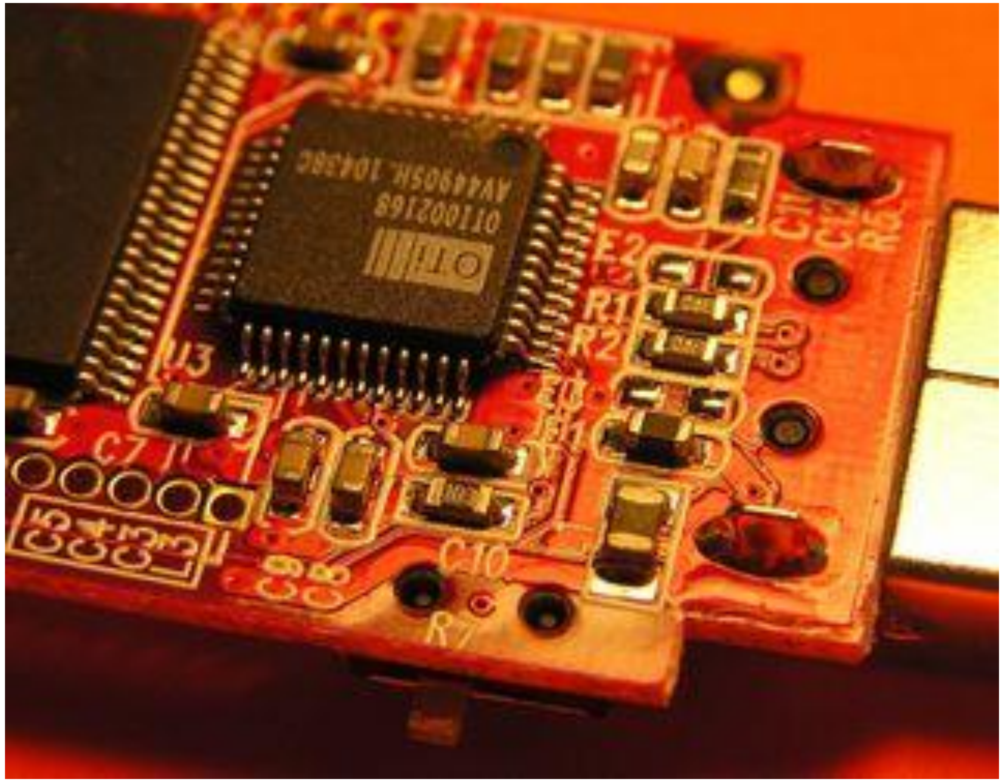
Vias

- Pad with a plated hole connecting traces from one layer of board to other layers.
- Attempt to minimize via use in your PCBs.
- Some component leads can be used as vias.



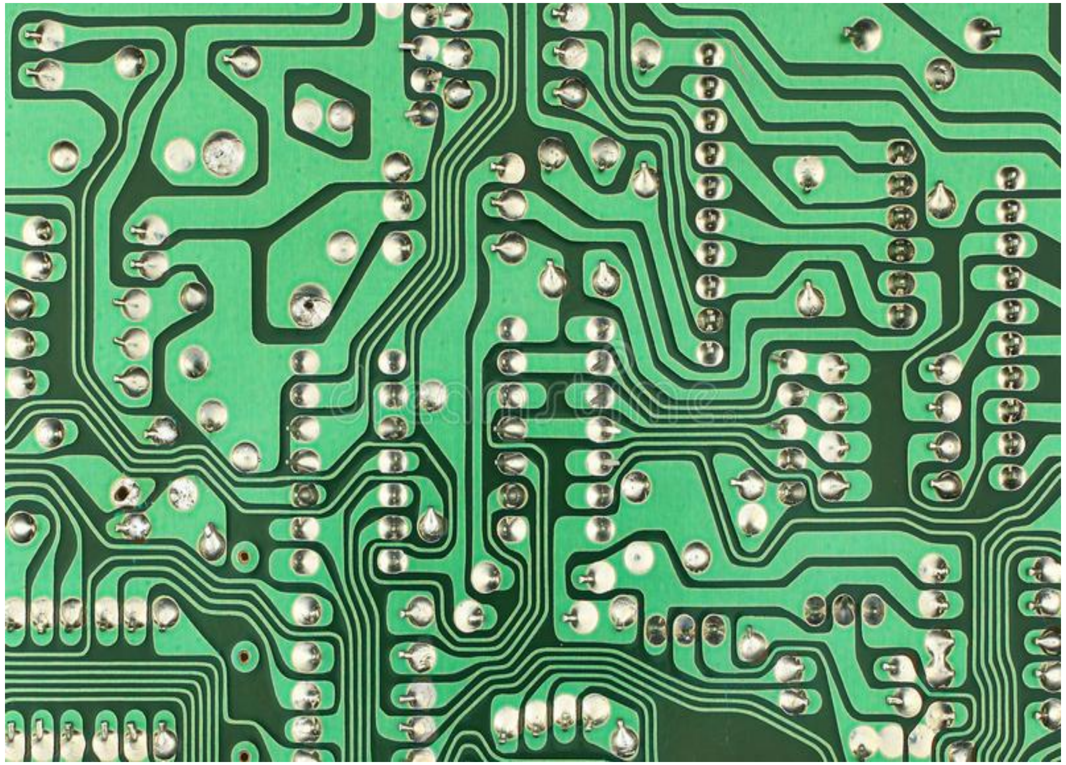
Top Metal Layer

- Most of the components reside on the top layer
- Fewer traces on the top layer
- Components are soldered to the pads on the top layer of PCB
- Higher circuit densities



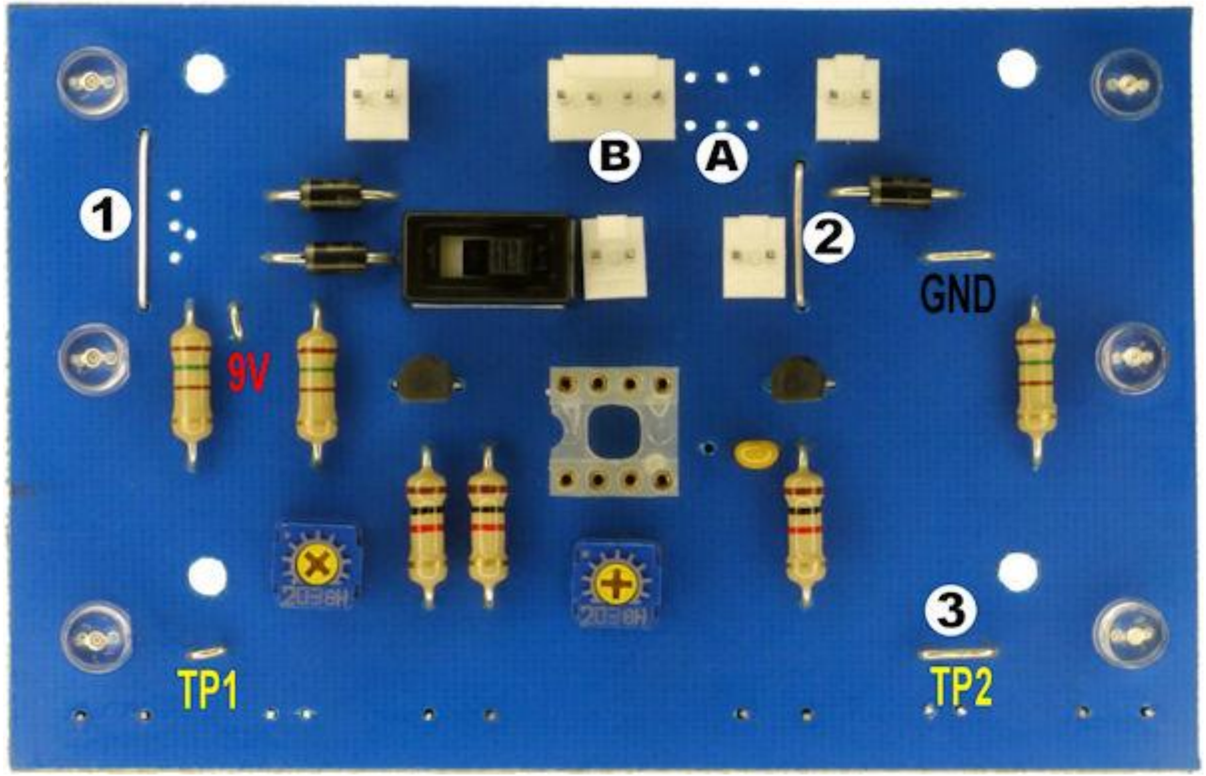
Bottom Metal Layer

- Few components on this layer.
- Many traces on this layer.
- Most soldering done on this layer.



Jumpers

- Often, many signal wires need to exist in too small of a space and must overlap.
- Running traces on different PCB layers is an option.
- Multilayer PCBs are often expensive.
- Solution: use jumpers



Physical Design Issues

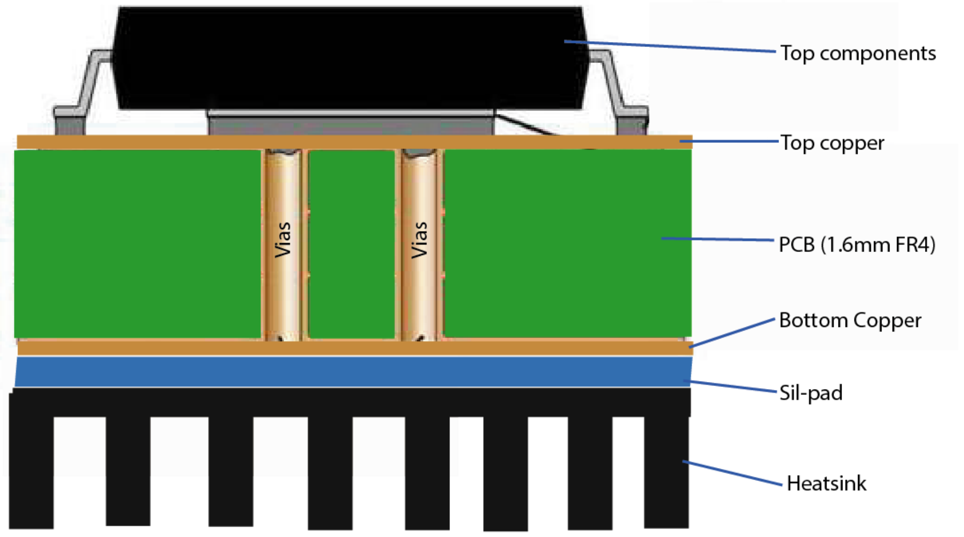
- Component Size
- Heat Dissipation
- Input and Output
- Mounting Points

Component Size

- Make sure components will actually fit.
- This especially applies for circuits that require high component densities.
- Some components come in multiple sizes. SMT vs Through Hole
- Sometimes you can get tall and narrow caps or short and wide capacitors.

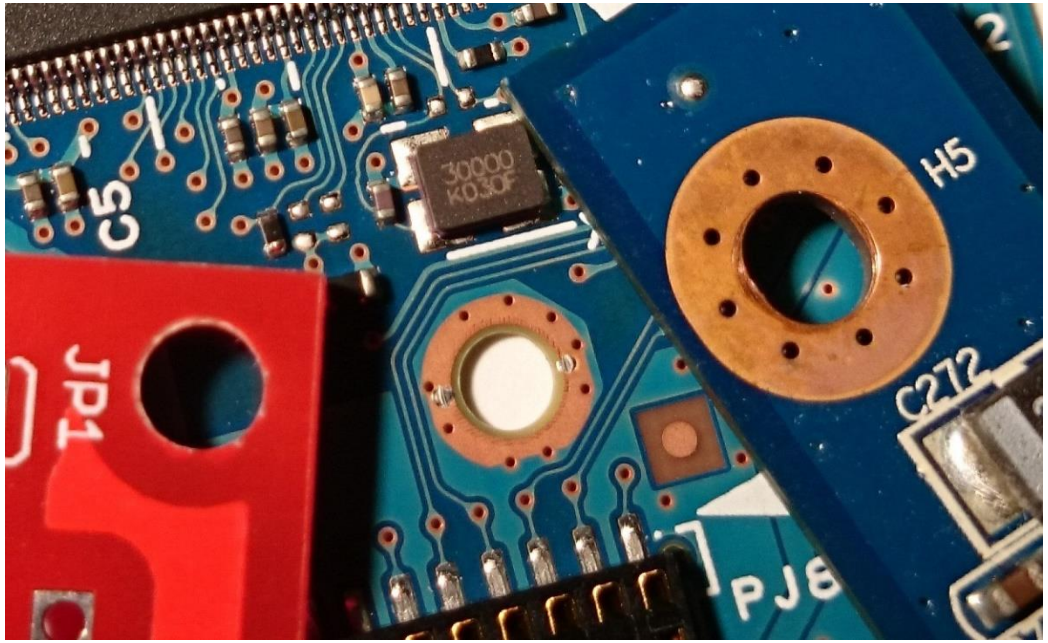
Heat Dissipation-Heat Sinks

- Heat sink dissipates heat off the component
- Doesn’t remove the heat just moves it
- Some components may get hot. Make sure you get a large enough heat sink.
- Data sheets specify the size of the heat sink
- A short circuit may result when two devices share the same heat sink



Mounting Points

- The PCB needs to be mechanically secured to something.
- Could be the chassis-consist of metal frame on which the circuit boards and other electronic components are mounted.
- Could be another PCB/socket on PCB.
- Could be attachments to a heat sink.



Pre-work

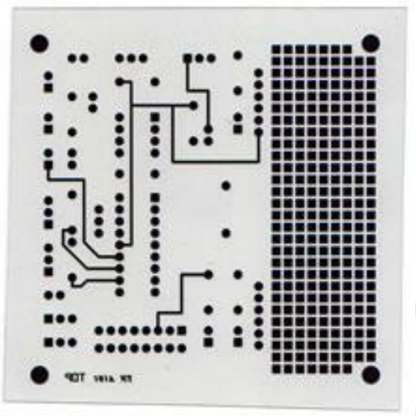
- Thoroughly simulate your circuit-make sure the circuit worked in simulations
- Thoroughly test the prototype-make sure the circuit worked on the bread board
- Have all the data sheets handy for every components
- Play around with the placement of the components

Simulations

- Important to simulate the circuits before building them
- Allow margin for component tolerances
- Avoid using precise components. e.g a PWM controller that requires exact 10 V DC to work and will fail if there is 10.01V
- High performance circuits or SMT devices require PCBs and should be simulated extensively first.

Steps in PCB design

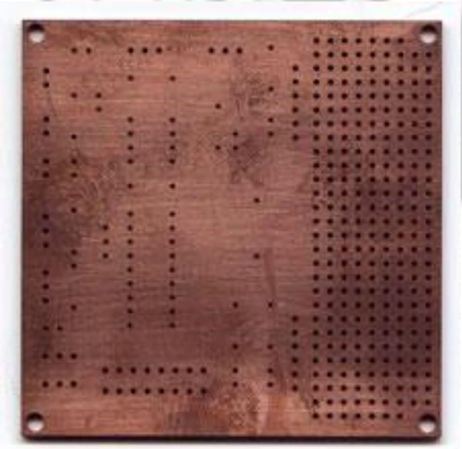
1. Film Generation



2. Shear Raw Material

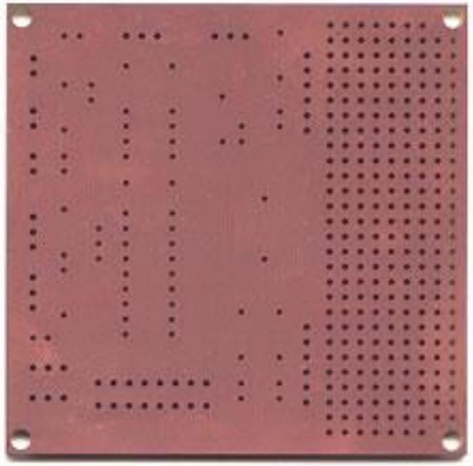
Industry standard 0.059" thick, copper clad, two sides

3. Drill Holes



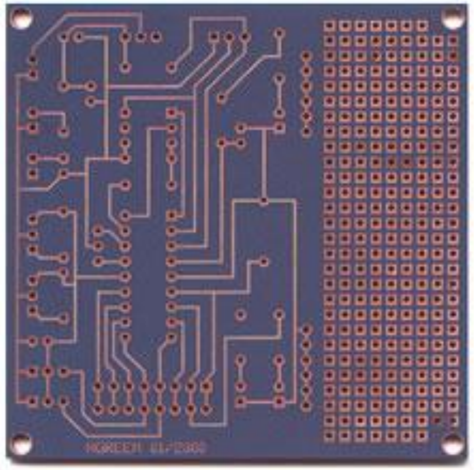
Steps in PCB design

4. Electrolus copper



Apply copper in hole barrels

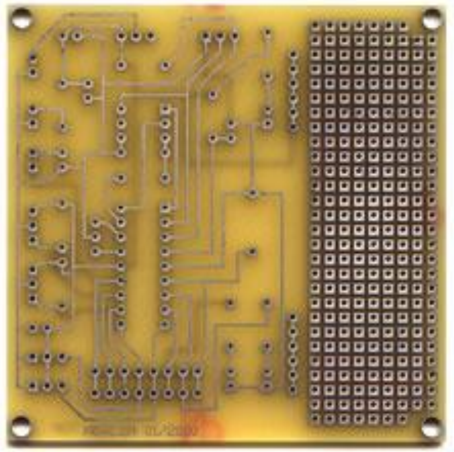
5. Apply Image



Apply Photosensitive Material to develop selected areas from panel

Steps in PCB design

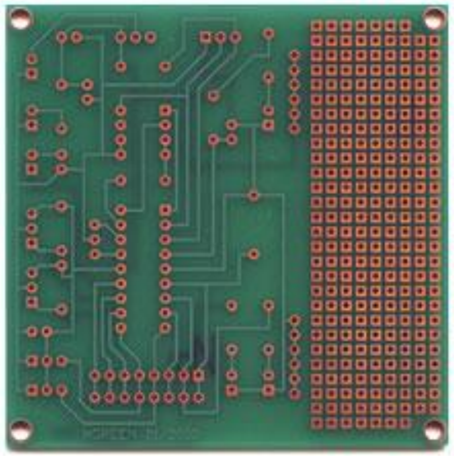
6. Strip and Etch



•Remove dryfilm, then etch exposed copper

•Tin protects the copper circuitry from being etched

7. Solder Mask



Apply solder mask area to entire board with the exception of solder pads